



**Verified Carbon
Standard**

KAMPONG CHHNANG 60 MW SOLAR POWER PROJECT



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Monitoring period	11-November-2022 to 31-December-2023
Crediting period	11-November-2022 to 10-November-2029
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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Kampong Chhnang 60 MW Solar Power Project, referred to as "the project," is a solar power plant developed by Prime Road Alternative (Cambodia) Co., Ltd., with a total capacity of 60 MWac. Located approximately 60 km northwest of Phnom Penh in the south of Kampong Chhnang province, Cambodia, the project can be accessed via provincial and rural roads. A partnership between the Asian Development Bank (ADB) and Electricite du Cambodge (EDC), Cambodia's national power utility, resulted in the development of a 100-megawatt (MW) National Solar Park. The first 60 MW (AC) of solar power under Phase I was awarded to Prime Road Alternative (Cambodia) Co., Ltd., in 2019 through a competitive reverse auction tender.

Cambodia's energy landscape is primarily dominated by coal and hydropower. In the baseline scenario, the country faces challenges such as heavy reliance on imported fossil fuels, power outages, and environmental impacts from hydropower and coal-fired plants. The Kampong Chhnang project, a solar power initiative, contributes to the national grid by supplying an estimated annual electricity of 141,719 MWh. This helps reduce dependence on imports, fossil fuels, enhances resilience to climate change, and supports Cambodia in meeting its Nationally Determined Contributions (NDCs).

Without the project, the baseline scenario involves generating an equivalent amount of electricity from grid-connected fossil fuel-dominated power plants. Therefore, the electricity delivered to the grid by the project would otherwise have been generated by these fossil fuel plants or new generation sources.

Construction of the project commenced on 13 August 2021, and was commissioned on 11 November 2022. The project owner has secured a long-term Power Purchase Agreement with Electricite Du Cambodge (EDC). On an average the project reduces approximately 81,868 tCO₂eq of GHG emissions annually. Over the first crediting period of seven years, the project results in an estimated emission reduction of 573,078tCO₂eq.

For the current monitoring period from 11 November 2022, to 31 December 2023 (inclusive of first and last date), the project has supplied 156,242 MWh of electricity to the national grid, resulting in an emission reduction of 91,839 tCO₂eq.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
<i>Joint Validation and verification</i>	11-11-2022–31-12-2023	VCS	4K Earth Science Private Limited	One year one month 20 days

1.3 Sectoral Scope and Project Type

Sectoral scope¹	1 - Energy (renewable/non-renewable)
Project activity type	Renewable Energy Project

This project is not an AFOLU project and is not a grouped project.

Sectoral scope	NA
AFOLU project category²	NA
Project activity type	NA

1.4 Project Eligibility

1.4.1 General eligibility

The Project is a 60 MW grid connected electricity generation activity using solar power in Cambodia, which is a Least developed Country (LDC)³ as designated by the United Nations. As per VCS Standard v4.7, section 2.1 -Table 1, grid connected solar power plants located in LDCs are within the VCS program scope, making this project eligible under VCS.

As per VCS Standard v4.7, section 3.8, the project start date is the date on which the project began generating GHG emission reductions. With operation starting on November 11, 2022, and considering subsection 3.8.1, the project is eligible to undergo validation within two years of its start date, i.e., before November 10, 2024, meeting the validation deadline requirements.

Developed under the CDM-approved methodology ACM0002 – Grid-connected electricity generation from renewable sources – Version 21.0⁴, this project aligns with VCS GHG program and methodology standards, qualifying it for participation in the VCS program.

1.4.2 AFOLU project eligibility

Not Applicable. This project is a solar power project which falls under the non-AFOLU project category.

1.4.3 Transfer project eligibility

Not Applicable as this project is not registered under any other GHG programs and is not seeking to transfer.

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

² See Appendix 1 of the VCS Standard

³ https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/publication/ldc_list.pdf

⁴ <https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL01OU2RYC6N3HASIXV7K84QBG9>

1.5 Project Design

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

Grouped Project Design

Not applicable as this project is not a grouped project

1.6 Project Proponent

Organization name	Prime Road Alternative (Cambodia) Co., Ltd.
Contact person	Mr. Rassa Herabat
Title	Head – Asset Management & Project Manager RE
Address	1 TP & T Tower, 22nd Floor, Soi Vibhavadi Rangsit 19, Vibhavadi Rangsit Road, Chatuchak, Chatuchak, Bangkok Thailand 10900
Telephone	+66 957936356
Email	rassa.h@primeroadgroup.com ; assetmanagement@primeroadgroup.com

1.7 Other Entities Involved in the Project

Organization name	Kosher Climate India Private Limited
Role in the project	Project Representative
Contact person	Vamsi Krishna M
Title	Director
Address	Zee Plaza, No.1678, 27th Main Rd, Near KLM Mall, Sector 2, HSR Layout, Bengaluru, Karnataka 560102, India
Telephone	+91 99453 43475
Email	vamsi@kosherclimate.com ; narendra@kosherclimate.com

1.8 Ownership

The project activity is being undertaken by Prime Road Alternative (Cambodia) Co., Ltd., owns all the legal rights and is the sole owner of the project activity. It has already obtained all the necessary permits/consents from the various concerned government departments that are required to develop and operate a solar power project in the Cambodia. Further, the tender award certificate, land lease agreement, Environmental Impact Assessment (EIA) license, equipment purchase contract and the power purchase agreements are the evidences for the ownership of the plant, equipment's and power generation. This is in line with the VCS Standard, v4.7, subsection 3.7.1 requirement 3.

1.9 Project Start Date

Project start date	11-November-2022
Justification	Project start date is the date on which project started generating electricity and hence began generating GHG emission reductions.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	11-November-2022 to 10-November-2029

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☒ < 300,000 tCO₂e/year (project)
☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
11-Nov-2022 to 31-Dec-2022	11,639
01-Jan-2023 to 31-Dec-2023	82,928
01-Jan-2024 to 31-Dec-2024	82,555

01-Jan-2025 to 31-Dec-2025	82,183
01-Jan-2026 to 31-Dec-2026	81,813
01-Jan-2027 to 31-Dec-2027	81,445
01-Jan-2028 to 31-Dec-2028	81,079
01-Jan-2029 to 10-Nov-2029	69,436
Total estimated ERRs during the first or fixed crediting period	573,078
Total number of years	7
Average annual ERRs	81,868

1.12 Description of the Project Activity

The project is a new greenfield photovoltaic facility with an installed power capacity of 60 MW, which is derived from 77.006 MWp of its total installed DC power capacity connected to the Electricite du Cambodge (EDC) substation within the National Solar Park (NSP). The Project comprises 300 units of 200kVA (1500V) Huawei String inverters combined with 10 sets of box-transformers. The PV Modules are installed on single horizontal axis tracking system, consisting of 2 types of tracking row configuration. One type has 3 strings of PV modules and the another with 2 strings of PV Modules, all installed in one row portrait orientation. The operational lifetime of major project equipment is estimated to be 20 years, The estimated energy generation by the plant in the first year is 141,719⁵ MWh which will go down considering an annual PV module degradation rate of 0.45%⁶.

There are total of 10 sub-systems, 9 of which have 6.5MVA box-transformers and the other one has a 3.25MVA box-transformer, each one connected to 4 circuits of 22kV switchgear side of the 115kV Project substation. The four (4) output of 22 kV ring main system (each comprising 20 MW output) will be connected to the Interconnection point at EDC's substation via underground cable (from the nearest box transformer to the Interconnection point). The Interconnection Point is at four (4) of 22 kV incoming feeders of EDC substation. The revenue meters are also installed at the Interconnection Point. These meters are of Landis+Gyr E650 make and have an accuracy class of 0.5s.

Incoming feeders, main and back-up meter details are as below:

1. Incoming Feeder #1, main meter No. 55548513, back-up meter No. 55548515
2. Incoming Feeder #2, main meter No. 55548514, back-up meter No. 55580597
3. Incoming Feeder #6, main meter No. 55548500, back-up meter No. 55580600
4. Incoming Feeder #7, main meter No. 55548498, back-up meter No. 55548501

Table I: Description of the facility

⁵ As per the third-party energy yield assessment provided in the technical due diligence report

⁶ As per the annual PV module degradation rate provided in the manufacturer datasheet

Nominal DC Power of the facility	77.00616MWp
Nominal power of PV module	540 Wp
Type of PV module (Photovoltaic Panel)	540W
Module Technology	Mono-PERC Bifacial PV module
No. of modules (Total) Mounting substructure	142,604 pcs Horizontal Single Axis Tracking system: East-West trackers
Inverter	300 sets of Huawei SUN2000 - 215KTL
Smart Transformer Station	9 sets of 6500 kVA 1 sets of 3250 kVA
Ring main unit (RMU)	10 sets of 22KV RMU (available in Smart Transformer station)

Table II: Summary of the technical specifications for the key components of the Project

Technology	Type	Manufacturer	Model
Solar PV Module	Monocrystalline bifacial	Suntech	STP540S-C72/Pmh+ (540Wp)
Inverter	String Inverter	Huawei	SUN2000-215TL-HO
Transformer	Box Transformer	Huawei	3250kVA
	Box Transformer	Huawei	6500kVA
Monitoring Equipment	Secondary standard	Kipp & Zonen	CMP10
	RTD sensor	OMEGA	SA2-RTD
	Weather transmitter	Vaisala	WXT536
Mounting Structure	Horizontal single axis tracker	Clenergy	EzTracker D1P

The objective of this facility is in line with the government's initiative to enhance power generation in the country, addressing the increasing demand and broadening the power sources. The solar photovoltaic facility is set to offer a clean and sustainable energy source, aiming to decrease the country's carbon footprint and support global endeavors against climate change. This initiative is a crucial stride toward achieving sustainable development in Cambodia, positively impacting the environment, economy, and society. The project seeks to reduce anthropogenic emissions of greenhouse gases (GHGs), estimated at an annual average of 81,868 tCO₂e, by displacing approximately 139,279 MWh/year of electricity produced by thermal fossil fuel-based power plants. Beyond emission reduction and diversification of power sources, the project will generate

employment opportunities, fostering the local economy. The installation of solar panels, inverters, transformers, and other essential equipment will involve the local workforce, contributing to the growth of the renewable energy sector in the country.

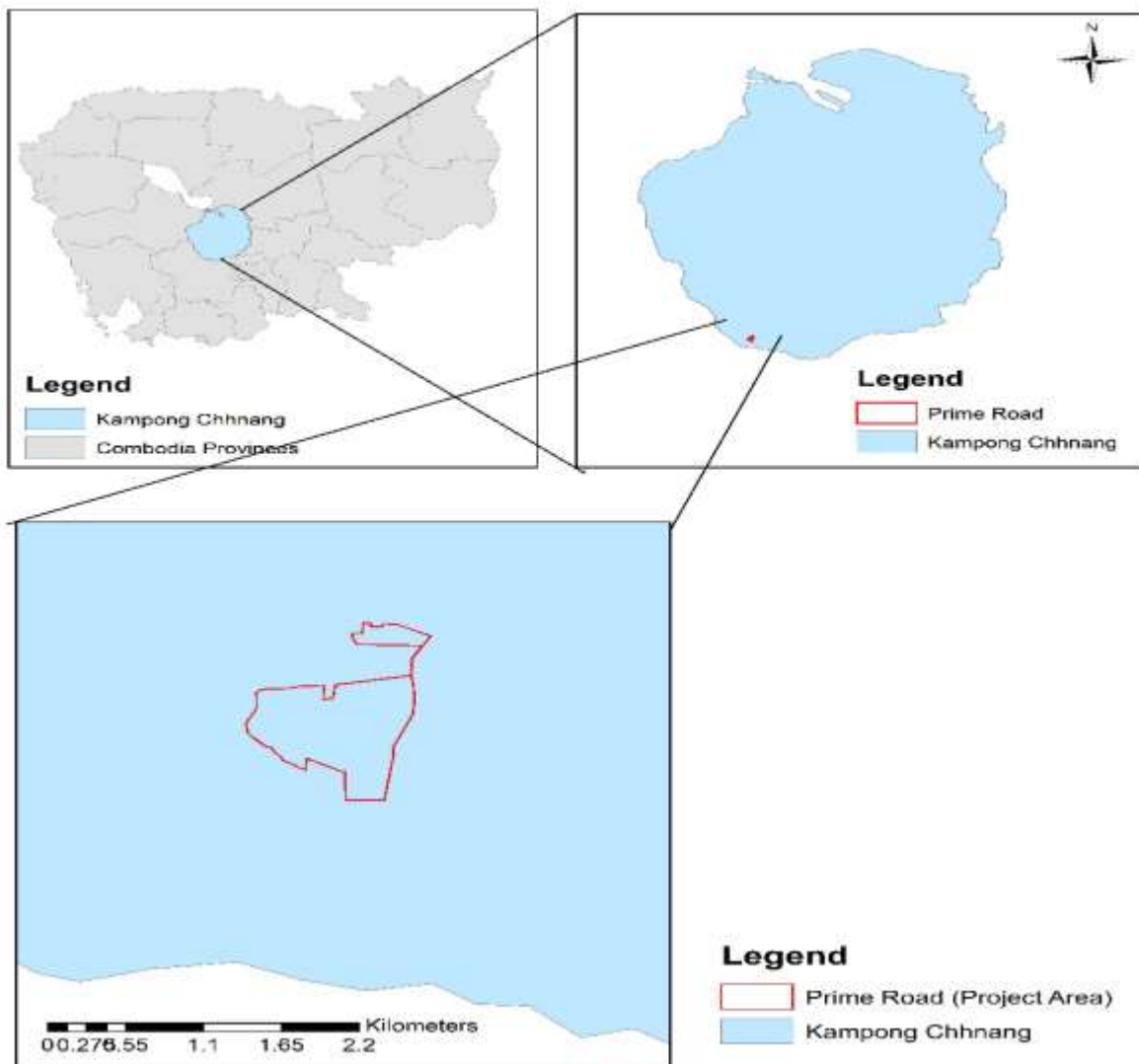
1.13 Project Location

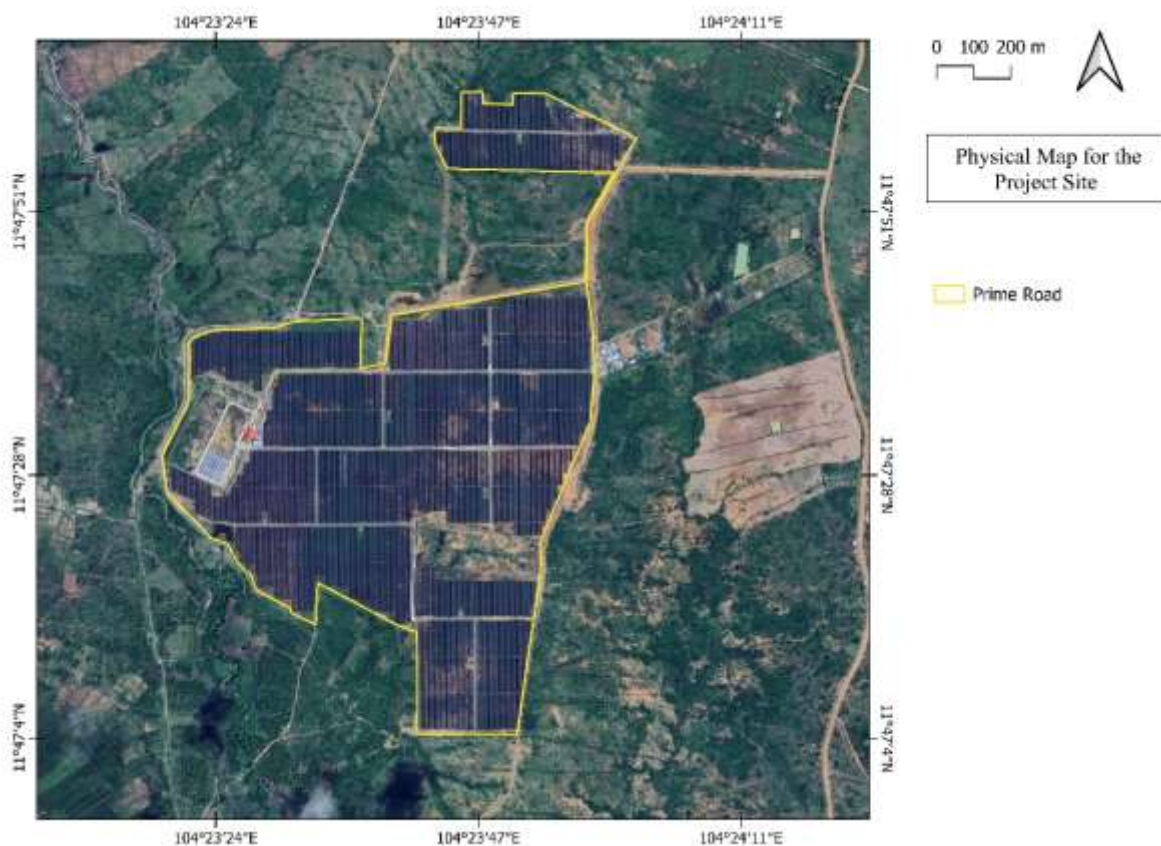
The 60MW solar plant covers an area of 97 ha within the planned 250 ha NSP and is located in Tuek Phos district of Kampong Chhnang province, Cambodia.

Table III: Location and description of site

Site Name	Longitude	Latitude	Size	Location
Kampong Chhnang	11°47'31"N	104°23'50"E	~97 hectares	Kampong Chhnang province, Tuek Phos, Kbal Tuek

Figure 1: Geographic Location of the project





1.14 Conditions Prior to Project Initiation

The project activity is the installation of a greenfield power plant and the baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

The conditions prior to the implementation of the project activity are same as that of the above-described baseline scenario. The project activity does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project is adhering to the relevant legal and regulatory guidelines of the Cambodian Power Sector, governed by the Electricity Law⁷ and the primary legislation overseeing Environmental Impact Assessment (EIA) in Cambodia, known as "The Law on Environmental Protection and Natural Resources Management⁸." This law outlines institutional responsibilities, mandates

⁷ https://cdc.gov.kh/wp-content/uploads/2022/04/LAW-ON-ELECTRICITY_010202-.pdf

⁸ https://cdc.gov.kh/wp-content/uploads/2022/04/Law-on-Environmental-Protection-and-Natural-Resource-Management_961224.pdf

impact assessment criteria, and specifies procedures for conducting environmental assessments.

The 60 MW solar power project has obtained all required approvals for its development and initiation from the appropriate authorities, ensuring alignment with local laws and regulations. Further, the project is aligned with the direction from Ministry of Mines and Energy which is to encourage exploration and development of environmentally and socially acceptable energy resources needed to supply all sectors of Cambodian economy.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

This 60 MW solar power project, located in Kampong Chhnang province, Cambodia, is a critical component of the National Solar Park initiative. The project contributes directly to Cambodia's emission reduction targets as outlined in the country's Nationally Determined Contribution (NDC), which aims to achieve a 21.3% reduction in emissions by 2030. In line with this goal, the project focuses on mitigating emissions from the electricity sector through the expansion of renewable energy power generation.

The project is designed to address three key national priorities: (i) expanding low-cost power generation to meet the growing demand for energy, (ii) diversifying the power generation mix to increase the share of clean energy, and (iii) implementing competitive tenders and other global best practices in the energy sector. By utilizing solar photovoltaic (PV) technology, the project directly replaces fossil fuel-based energy sources, thereby reducing greenhouse gas (GHG) emissions and contributing to climate change mitigation.

In addition to its environmental benefits, the project contributes to the advancement of Sustainable Development Goal (SDG) 7 by ensuring access to affordable, reliable, sustainable, and modern energy for all. The increased availability of clean energy is expected to drive sustainable economic growth, enhance energy security, and improve equity by providing affordable power to underserved communities.

The project aligns with Cambodia's nationally stated sustainable development priorities and the country's long-term vision for transitioning to a low-carbon economy.

Table IV: Contributions of the project activity to SDGs

	By generating an estimated average annual output of 139,279 MWh of renewable energy, the project significantly contributes to the realization of SDG 7, which strives to ensure access to affordable, reliable, sustainable, and modern energy for all. This contribution aligns with the overarching goal of fostering a greener and more sustainable future.
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	Throughout the construction and operational phases, the project has actively created numerous direct and indirect employment opportunities, predominantly benefitting the local population in the region. This proactive approach contributes to the fulfilment of SDG 8, which seeks to promote sustained, inclusive, and sustainable economic growth, along with full and productive employment and decent work for all.
	Through its efforts in emission reductions, the project aligns with the objectives of SDG 13, which calls for urgent action to combat climate change and its adverse impacts. By taking substantial measures to decrease greenhouse gas emissions, the project actively contributes to the global initiative of safeguarding the planet for the well-being of future generations.

1.18.2 Sustainable Development Contributions Activity Monitoring

The Kampong Chhnang solar power project generates electricity using renewable solar energy. With an estimated annual generation of 139,279 MWh of renewable energy, the project activity contributes to achieving SDG 7. Further, through emission reductions, this effort aligns with SDG 13 which calls for urgent action to combat climate change and its impacts. During the current monitoring period, the project proponents have conducted training sessions to educate and build capacity in fire safety, environmental & social protection, O&M related trainings and training on grievance mechanism in place. 42 individuals have been successfully employed, contributing to the realization of SDG 8, which aims to promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

Table 1: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
Sequential row number	SDG Target number	Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator	Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)	Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.	Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	In the reported monitoring period (11/11/22 to 31/12/23), the net renewable energy supplied to the grid is 156,242 MWh. Please refer section 7.1 for more information	Until the end of the current monitoring period the project has generated 156,242 MWh of renewable energy and the project is expected to increase the share of renewable energy in the national grid by 3,124,840MWh over its project lifetime

2)	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities	Implemented increase	activities to	Project activity supports creation of short-term and long-term job opportunities for men and women during the construction and operation of the project activity. During the current monitoring period (11/11/22 to 31/12/23), the project has created new employment and generates income for 42 people including men and a woman. Please refer appendix 2 for more information	By the end of this monitoring period the project activity has resulted in generating job opportunities and income generation to 42 people.
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented decrease	activities to	By generating 156,242 MWh of renewable electricity the project activity has avoided 91,839 tonnes of carbon emissions into the atmosphere during the current monitoring period. Please refer section 7.5 for more information	Until the end of the current monitoring period the project has avoided 91,839 tonnes of carbon emissions and during the project lifetime it will avoid the release of approximately 1,836,780tonnes of carbon into the atmosphere

1.19 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project activity is not an AFOLU project

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable, as this project activity does not involve any additional information that may have a bearing on the eligibility of the project, the GHG emission reductions, or the quantification of the project's reductions.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

Stakeholders include various individuals, groups, and institutions who have an interest in, could be impacted by, possess the ability to significantly influence, or are crucial for achieving the desired project outcome. The list of relevant stakeholders identified through stakeholder mapping includes:

- Local people, communities, and representatives expected to be directly or indirectly affected by or interested in the project.
- Local policymakers and representatives of local authorities.
- Local non-governmental organizations (NGOs) and Women Groups focusing on topics relevant to the project or collaborating with communities likely to be affected by the project.

Legal or customary tenure/access rights	This project is a component of Electricite du Cambodge's (EDC) National Solar Park (NSP) initiative. The project proponent has secured a 97-hectare lease for a 60 MW photovoltaic (PV) plant from EDC. EDC acquired the land through commercial negotiations, following a willing buyer-willing seller approach. As a result, the project site does not involve any territories or resources to which the stakeholders have customary rights or access.
Stakeholder diversity and changes over time	The plant is situated in Prey Chrov village in Kbal Toeuk Commune, a socio-economic survey was carried out in Kbal Toeuk commune and 2 other nearby communes – Chhean Loerng Commune and Prambei Mom commune to determine relevant demographic characteristics of the communities. Based on the socio-economic profile, stakeholder mapping was carried out to ensure that equitable, non-discriminatory and meaningful consultation was carried out. The stakeholder group exhibited notable diversity across social, economic and cultural dimensions. Socially, individuals from various backgrounds, including different age groups, ethnicities and socio-economic classes were present. It was ensured that women, disabled or elderly people, informal sector workers, poor communities and stakeholders with different financial capacities were represented and involved. The diverse stakeholder groups interacted in multifaceted ways, driven by their unique perspectives and interests to voice out their opinions. Overtime, the composition of these groups may undergo changes due to evolving circumstances or project related developments. New individuals or entities may join, a few might disengage. To monitor these changes and to ensure that the project continues to align with the diverse needs and expectations of its stakeholders, the project proponents will engage with the stakeholders every six months and has implemented a Grievance Redressal Mechanism (GRM).
Expected changes in well-being	The socio-economic survey revealed that in the baseline scenario there are no ethnic groups apart from Khmer and majority of the households in the project area practice farming while some households run small businesses. Livelihood generation and employment on plantations are reported to be temporary and seasonal. Agriculture is the main occupation with secondary income generating

	activities being livestock rearing, fishery and working in factories, where both men and women are involved. Poverty is a concern in the survey area due to limited job opportunities. This project activity will employ local populace resulting in economic and livelihood generation opportunities during the construction and operation phase. Further, there are no protected areas or habitats of particular biodiversity value within the project area of influence. The solar plant site is located in modified environment (disturbed by human activity) dominated by scrubland with scattered trees and paddy fields hence no changes to the ecosystem services important to the stakeholders have been identified.
Location of stakeholders	The locations of the stakeholders are – Kbal Toeuk commune in Toeuk Phos district, Chhean Loerng commune in Samaki Meanchey district, Kampong Chhnang province and Prambei Mom commune in Kampong Speu province. The solar plant is situated within the administrative are of Prey Chrov Village in Kball Toeuk Commune, where the population is located around 18km from the commune where most of the stakeholders live and is unlikely to experience any direct or indirect impacts from the project activity. Further, there are no habitations/settlements within the 60 MW solar plant or private ownership with formal land titles or traditional land holding. The project’s transportation is partly through Prambei Mom commune, the closest human receptors that will be likely affected due to transportation are those living along the existing access road. Potential impacts due to transportation issues such as dust and noise have been assessed based on Cambodian and International environmental standards and mitigation measures were developed to help minimize any direct or indirect adverse effects associated. Also, the transportation need of the project was minimal since the total planned construction was only 18 months and the needs of transporting materials in and out of the project site were not on a regular basis.

Location of resources	There are no habitations/settlements within the project site. This site was previously used for cassava plantations and was later reportedly abandoned. The site assessment conducted by the project proponent for 197 ha identified the site as mainly scrubland and paddy fields. The site surrounding the project, is now a grass-field and a grazing area for local villagers. This is a kind of opportunistic grazing by villagers. There are no territories or resources in the project site which are owned by the stakeholders or have customary access to.
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2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	The project proponent conducted the stakeholder consultation on July 21, 2022, during the construction period to provide updates on project progress, assess its impact on the community, and offer stakeholders an opportunity to voice their concerns, share recommendations for improvement, and provide feedback.
Stakeholder engagement process	Key stakeholders were identified by stakeholder mapping considering social, cultural and gender-specific factors that might influence their engagement. Representation and participation from diverse groups such as local communities, local governments, national & local NGOs, private sector entities comprising women, men, elderly, girls and boys etc. was sought by inviting them to attend the meeting through: • Personal face to face invitations to government officer/NGOs • Personal face to face invitations and Invitation via phone calls to the villagers Local people were invited to the meeting via personal face to face invitations and phone calls. Local government officers were invited to the meeting via face-to-face invitations. The meeting announcements were made well in advance after consulting with the commune and village representatives. The agenda of the meeting, the Project

	Information Booklet (PIB) and guide questionnaires for Focus Group Discussions (FGDs) with men and women in Khmer language were distributed well in advance. The consultation was undertaken in both village and provincial levels. It was a two-way information sharing channel where participants provided their comments and feedbacks. The results of the consultations; concerns raised & recommendations made have been addressed in the ESIA & Environmental and Social Management Plan (ESMP) of this project activity. Furthermore, the project proponents have established a Grievance Redressal Mechanism (GRM) which provides an accessible platform for receiving and facilitating resolution of grievances from affected stakeholders.
Consultation outcome	Public participation as part of the consultation included information disclosure via distribution of project information booklets (PIB) in affected communes during consultations, public meetings at district, and village level and focus group discussions (FGDs) with both women and men of the communes. Furthermore, the project proponent has ensured consistency in social and environmental monitoring in compliance with the local laws, ADB safeguard policy statement (SPS) and IFC Performance Standard requirements. As required by the Cambodian regulation, the project proponent will be contributing USD 1,000/Year towards environmental and social fund. In total 29 local people including local men and women, village chiefs were part of the meeting. The key points discussed with each of these stakeholders are provided below: Stakeholder Category - Local community Discussed Key Points: • Providing the project progress update • Are there any impacts caused by the project? • Provide an opportunity for relevant stakeholders to raise their concerns and recommendation.

- Any request for project to improve?

Outcomes in brief

- Participant 1: As the project uses this road for the transportation, please drive the vehicles safely and slowly and avoid certain time (sleeping or school times).
- Participant 2: My family also go to project site for working as well and I'm happy that the project developed in this area.
- Participant3: The project has no big concern with us because we don't own any land there.
- Participant 3: As there is not much transportations moving on the road (the road seem quiet traffic), but most of the kids they playing near/on the road as well. So please drive slowly and safely (especially the transportation truck to the site).

Stakeholder Category - Royeas Village Chief

Discussed Key Points

- Providing the project progress update
- Are there any impacts caused by the project?
- Provide an opportunity for relevant stakeholders to raise their concerns and recommendation.
- Any request for project to improve?

Outcomes in brief

- Because the project located in Prey Chrov village, there is no land conflict on the local people in this village, but I would like the project to take action on any grievance from villagers as soon as possible if occurred.
- I would also request for a proper caution in regard of traffic, especially traffic accidents resulting from transporting materials and workers during project develop phase.

Stakeholder Category -First Deputy of Chean Lerng Commune

Discussed Key Points

- Providing the project progress update
- Are there any impacts caused by the project?
- Provide an opportunity for relevant stakeholders to raise their concerns and recommendation.
- Any request for project to improve?

Outcomes in brief

- I have no comment or request with the project since the village chief and community people have raised many ideas regarding to this.
- I'm happy that the project developed near to our area even though the project itself is not in our commune.
- I'm happy to attend any other meeting in the future if the project requests me to join.

There are some requests and questions from villagers and stakeholders as below:

Questions from the stakeholders:

1. Up to this time, are there any traffic accidents happened caused by project's workers or employees?

PP: There are no traffic accidents caused by project.

2. How many families that working for the project?

PP: There are a lot of family working for the project and hope that there is another project to develop in this community in the future.

3. Talking about inflation, by comparing from the state that the project hasn't exist and now. How is it different? The most concerning from us is gasoline and the price of other things also rise up a little bit as well.

PP-PRAC: The price of gasoline and any other things rise up are not the fault of project. It because of the economic war between Russia and another north country. And of course,

if the gasoline price rises up, the price of other things would rise up as well (transportation cost goes up).

Questions from community people:

1. Does the project can be affected to the living and farming of the community people or not?

PP: No, it is not. The project is clean energy source, it means that there is no effect to the living (health) or your farming (vegetables). Apart from that, for safety, we also have fence to cover all over the project area, so we are sure that your animals or human cannot go inside without permission from the security.

2. Does the solar farm can be electricity shock to human or our cattle?

PP: Yes, it can. But only one reason that you or your cattle go inside our project area without permission by security guard. If you or your cattle just stay outside the project area (Outside the fence), we are sure that there is no problem happen to you.

3. Will the electricity that generated provide to community people for using or not?

PP: Yes, it the same as the electricity that you are using nowadays, the benefit that our project can provide to you is that after the project come to COD, the power generated will connect to EDC sub-station and EDC will provide to you next. Then the price of electricity in KWh will be cheaper than before if possible.

Requests from community people:

1. The truck for transport materials drives so fast, please tell them to slow down as the people and kids use this road also. We afraid that there would be traffic accidents happen cause of those trucks in the future.

PP: Agreed

2. In this season, we cannot growth rice. Please let us know in case the project need more people to work.

PP: Agreed

	All of participants supported the project. Main environmental concerns expressed by the participants were related to increase in dust in the dry season, construction debris dumping during construction and the operation stage. Measures to address these concerns have been integrated in the design and ESMP. Note: Evidence of Stakeholder Consultation is attached in Appendix 3.
Ongoing communication	The Project Management Office (PMO) has established a Social and Environmental Compliance Unit (SECU) for handling complaints/grievances attributed to social or environmental issues based on several discussions and meetings with ADB/IFC. Further, there is a Grievance Redress Mechanism (GRM) to receive and facilitate resolution of project affected persons' concerns, complaints, and grievances about the project's environmental and social performance. GRM involves a Grievance Redress Committee (GRC), which comprises of the relevant commune and/or village chiefs, provincial officers or their nominee, District Officer from the Cadastral office or their nominee, Contractor and a witness of the affected person; at least one person in the GRC will be female. The GRC is expected to: (i) resolve issues on land acquisition (if any), (ii) resolve issues on dust, noise, vibration, construction related nuisances to public, etc.; (iii) convene once a month to review complaints lodged (if any); (iv) record the grievances and resolve the issues within 30 days from the date the grievance was filed with the SECU; and (v) report to the complainant(s)/affected persons the status of grievance resolution and the decisions made or action taken. GRC has been established at provincial level ("Provincial GRC") with representation from commune councils across the project area of influence where the project components will be implemented. The key contacts for the GRC were handed over to the stakeholders in person during the Local Stakeholder Consultations.

Stakeholder input	During the consultation, the project proponents updated stakeholders on the project's progress, discussed potential impacts, and gave stakeholders the chance to express concerns and suggestions for improvement. Stakeholders expressed satisfaction with the project outcomes but requested more job opportunities due to their involvement in seasonal agricultural activities. The project proponents assured them of future employment announcements for local hiring. Concerns about transportation-related issues, including dust, noise, and potential accidents, were addressed. The project team emphasized their commitment to safety, highlighted the absence of transportation accidents, and assured continuous monitoring of noise and air pollution, which remained below harmful levels. Stakeholder comments mainly sought clarification, and as sufficient information was provided during the meeting, no changes to the project design are deemed necessary.
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2.1.3 Free Prior and Informed Consent

Obtaining consent	Consent for implementing project activities involved engaging with stakeholders since July 2018, when EDC initiated consultations before acquiring the land. Initial consultations with relevant stakeholders potentially affected were conducted by EDC's consultants in May and July 2018 to inform them of the proposed project, potential temporary impacts and project benefits. Furthermore, during the preparation of the ESIA report for 60MW solar plant, two public consultations at village level were organized on 27 September 2020 to disseminate the 60MW solar plant information and discuss the concerns and suggestions of the potentially affected peoples. Therefore, the periods May–July 2018 and 27 September 2020 can be considered as key milestones in the project
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development process when stakeholder consent was obtained before proceeding to the subsequent construction phase.

Total of 26 persons (11 females) participated in the FGDs at the village level. Participants were allowed to provide comments and feedback on the presentation. Few people turned in with some concerns and suggestions:

- Participant 1: Since the project will use this road for the transportation, please drive the vehicles safely, and avoid certain time (sleeping or school times) to not disturb the people and their local economic activities.
- Participant 2: If the project needs some labor, please consider selecting us in the local village so that we can get a job and some little income.
- Participant 3: The project must consult with villagers in advance if they place the transmission poles and pay a proper compensation.
- Participant 4: The project has no big concern for us. We do not own any land there.
- Participant 5: We do not concern much about the project impacts since it is Toeuk Phos district. What I concern is the placing the transmission poles to GS6. Just inform us and pay us the compensation before the project starts.
- Participant 6: The access road is cutting through the private land and plantation. The project must consult with the owners and address the land prices prior to building the route.
- Participant 7: We are happy to see the project and expect of job employment for our local villagers.

In response, it was confirmed that the identified impacts were limited to the construction phase of the solar plant project. Additionally, continuous monitoring of air quality and potential noise impacts on communities along transportation and access routes will be conducted, with appropriate mitigation measures developed as needed. Regarding employment, the project will create job

opportunities for local villagers, contributing to community development.

Additionally, consultations were carried out with the commune chief's and their feedback is as below:

PramBei Mom Commune Chief, Thpong District, Kampong Speu Province

- Because the project locates in Toeuk Phos district, there is no land conflict on the local people in the commune, but I would like to request for appropriate resolutions if there are conflicts occurred.
- I would also request for a proper caution in regard of traffic and public safety, especially traffic accidents resulting from transporting materials and workers during the project development phases.
- I suggest that prior to placing the transmission poles from the project, the project owner must consult with local authorities and the affected villagers to avoid or minimize land conflicts on private properties of the local villagers.
- I support the project.

Chhean Loueng Commune Chief, Toeuk Phos District, Kampong Chhnang Province

- Since the past this area belongs to private individuals and then they sold them to Mother Barang (local nickname) who bought all land in the area for cassava plantation and later the area is abandoned due to low productivity.
- The project site locates further away from the villages (Royas, Chhoun Chet) and the areas are privately owned. As EDC already purchased the land from owners, then there will be no serious land conflicts.
- I think the project activities will not cause any risks or damages on forest and wildlife resources because there are less of such resources and it is the former farmland.
- I suggest the project owner (1) comply with regulations especially the regulations on transporting materials and staff in and out of the site that would disturb or affect the

local villagers living along the road; and (2) cooperate with local authorities for every activity.

Mr. Soth Sang, Deputy-District Governor of Toeuk Phos

- The project owner must consult with local authorities and villagers in advance before the start of the project work and must ensure that there will be no conflicts on the right of way in front of their houses.
- The project owner must show the master plan and design of the project facilities and locations so that people are aware of the project and are ready to cooperate in case of there are conflicts or impacts from the project.
- I am very much happy with the development project because there is no sufficient power in Cambodia yet.
- I request for a safe transportation of materials and staff in and out of the project site that would cause traffic accidents unintentionally. Drivers must be trained and driving must be monitored regularly to minimize the impacts.
- The project must not fill the cracks, canals, or streams within and closed to the project site to avoid flood in the raining season. The project must prepare the drainage system to avoid flood and drain the wastewater from the project site.
- I think this project has no severe concerns on the environmental and social resources and the local villagers.

Additionally, a consultative meeting was held in Kampong Chhnang province on October 19, 2020. Chaired by Mr. Khem Channorn, the consultation meeting followed the agenda: (1) introduction to all participants, (2) remark by the chair, (3) project and ESIA presentation, (4) questions and answers (including feedback), and (5) wrap up and closure.

There were 20 participants from provincial departments and local NGO (Build Community Voice)

Mr. Khem Channorn – Deputy-Director

- It is a great pleasure to welcome the project since Kampong Chhnang province was very attractive to many development projects.

- This is not the only one solar plant. In this province there are some others including the one in Taing Krasaing commune.

- The solar plant is good to produce more energy and this energy is renewable. This means that we can generate electricity for a long time since the sun never runs out of its energy. This project can produce up to 60 MW and hope that this energy will help reduce our power scarcity and reliance on fuel source.

- We meet to today to welcome the project and learn about it and provide our inputs or comments to assure that the project minimizes the risks as predicted and brings more benefits than losses to our communities and Cambodia as a whole.

- I would like to invite all of you to look through the presentation and materials provided in this meeting and bring about the concerns and suggestions so that we develop the project in a manner that is environmentally friendly and sound.

Mr. Van Rady, inter-sectoral officer of provincial office

- I fully support the project. I would request that if all land plots are not yet properly purchased, please procedure with the buying-and-selling principles to assure it legality.

- I suggest having all plots combined into one-single tenure and registered with Land Department in the province.

Mr. Saing Sokhern – Deputy-governor of Toeuk Phos district

- There is no route from Toeuk Phos to the project site. Thus the project has to go through the national road no. 136 in Chhean Loeung commune. As far as I know, this project closes to Kiri Aphivat village, which is a newly-established village in that area, which borders with Toeuk Phos and Samaki Meanchey districts, and also closed to the Tbeng Srong community forestry. Most disturbed people are from

this new village think. Therefore, the project may need to build new road through Samaki Meanchey district. My request is to help build such road in this district for not only the benefit of the project, but also local people enjoy traveling around too. This road is about 4 kilometers.

- Because it is a government supported project, I totally support it.

Mr. Bong Bonny – Deputy-Director of Cults Department

- There is no impact on any sacred site as there is none of these in and around the project site.
- I fully support the project.

Ms. Phin Phalla – Deputy-Director of Women Affairs Department

- Transportation of materials and staff or workers must lower the risks of dust and minimizes the traffic accidents. Please water the road regularly to keep dust down where dust is potentially generated.
- Local people are not using the road properly. Please ask your driver to be more moderate and professional in driving i.e. with speed limits. Also, the project shall inform the local authority and people of the transportation schedule so that they could manage themselves to avoid any road challenges.

Mr. Nou Sokhon – Deputy-Director of MLV Department

- What is the labor required for this project during the construction phase? How many Cambodians and foreigners?
- I would suggest the project implement the safety measures for all workers and staff in this project. Also please prepare the workbook of all staff and workers and inform the MLV department prior to your groundwork.

Mr. Ouk Sophea – Deputy-Chief of Office of Transport and Public Work Department

- I support the project and the project is just aware of heavy transport to avoid road pressure and damages.

	• In case of heavy transport, please consult with the Department prior to any work. Mr. Saing Sokhern – Deputy-Cheif of Toeuk Phos District • I suggest the project inform us as local authority at least one week prior to any groundwork so that I can arrange local guard and people to accompany to avoid any unplanned conflicts. • Also please keep the local authorities alerted and informed of any work of the project Note: Evidence of Stakeholder Consultation is attached in Appendix 3. All leased land was obtained through commercial terms, following a willing seller-willing buyer approach. The project proponents fostered open dialogues, providing stakeholders ample opportunities to express concerns, ask questions, and share input. All raised concerns and inputs have been documented, and a grievance resolution mechanism is in place to address ongoing conflicts, demonstrating the project's commitment to prioritizing resolution and community harmony. Regular stakeholder engagement will persist throughout the project lifecycle to address evolving concerns and ensure ongoing support. Through this comprehensive process, the project not only obtained consent but also built a foundation of trust and cooperation with stakeholders, resulting in no ongoing or unresolved conflicts.
Outcome of FPIC	The result of the FPIC and stakeholder consultation is positive, with stakeholders expressing their support for the project. Additionally, a third-party assessment was conducted to confirm that the land acquisition adhered to the ADB SPS (2009) requirements. The verification affirmed that EDC acquired the land through commercial transactions, ensuring that the project did not encroach on land, relocate people without consent, or cause forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

Development process	Grievance Redress Mechanism (GRM) has been established in a systematic and culturally sensitive manner to address concerns raised by stakeholders. This involves a Grievance Redress Committee (GRC), consisting of relevant commune and/or village chiefs, provincial officers or their nominees, a District Officer from the Cadastral office or their nominee, the Contractor, and a witness of the affected person, with at least one female member. The GRC is tasked with (i) resolving land acquisition issues (if any), (ii) addressing concerns related to dust, noise, vibration, and construction-related nuisances to the public, (iii) convening monthly to review lodged complaints (if any), (iv) recording grievances and resolving them within 30 days from filing, and (v) reporting the status of grievance resolution and decisions/actions taken to the complainant(s)/affected persons. The Provincial GRC, established at the provincial level, includes representation from commune councils across the project area of influence where project components will be implemented. The key contacts for the GRC were handed over to the stakeholders in person during the Local Stakeholder Consultations, to ensure accessibility and awareness.
Grievance redress procedure	The Grievance Redress Mechanism (GRM) comprises the following conflict resolution procedure: Step 1: Affected persons/complainants can present complaints to the commune council through the village or commune chief, either verbally or in writing. The chief will provide an immediate written receipt and forward the complaint to the Social and Environmental Compliance Unit (SECU) of the project. The SECU will resolve the issue within one week through negotiation. Records of complaints and

resolutions will be submitted to the Project Management Office (PMO) as part of monthly progress reports.

Step 2: If the complainant is dissatisfied with the contractor's action or decision, they will take the issue to the PMO via the commune council. Grievances will be recorded in writing and forwarded to the SECU. The project proponent has 15 days to resolve the complaint through negotiation. If unresolved within 15 days or if the complainant remains dissatisfied, they can escalate the issue to the District office.

Step 3: The District office has 15 days to negotiate the complaint and attempt resolution. If satisfactory resolution is not achieved, the District office will refer the issue to the Provincial Grievance Redress Committee (GRC).

Step 4: The Provincial GRC will meet the complainant and attempt to resolve the issue. Within 30 days of complaint submission, the GRC must make a decision and inform both the complainant and SECU in writing.

Step 5: If dissatisfied with the Provincial GRC or its decision, the complainant can bring the case to the Provincial Court. The Court's written decision will be submitted to executing and implementing agencies. If still unsatisfied, the complainant can escalate the case to a higher-level court.

Step 6: If disputes remain unresolved or unsatisfactory, affected persons can directly discuss concerns with ADB's Energy Division, Southeast Asia Department, or through ADB Cambodia Resident Mission (CARM) and IFC's ombudsman office. If still dissatisfied, they can contact the ADB Office of the Special Project Facilitator through the accountability mechanism.

Recordkeeping and Reporting: The SECU will maintain records of grievances, including complainant details, complaint date, nature of grievance, corrective actions, and resolution dates. Summaries of filed and resolved grievances will be reported in quarterly project progress reports and semi-annual safeguard reports. The outcomes will be displayed/disclosed in project field offices,

	Commune council offices, and the Provincial Office if required.
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Grievances received	Resolution and outcome
NA	No grievance has been raised during the current monitoring period. The key contacts for the Grievance Redress Committee have been provided to the stakeholders during the Public Consultation Meetings so that these contacts are easily available to the stakeholders to raise their concerns or suggestions if any during the course of the project activity.

2.1.5 Public Comments

Comments received will be summarized in the table below once the project opens for public comments

Comments received	Actions taken
No comments received during the 30 days public commenting period.	No actions required

2.2 Risks to Stakeholders and the Environment

	Risks identified	Mitigation or preventative measure taken
Risks to stakeholder participation	No risk identified	No risks to stakeholder participation were identified during the preparation for stakeholder consultation. The project developer analyzed relevant stakeholders, risks, opportunities, and strategies for conducting the most effective consultation. To ensure inclusivity, the consultation venues were chosen centrally and made accessible to people of all castes, creeds, and sects in the society,

		ensuring that the location was considered safe by everyone in the community. Focus Group Discussions were conducted specifically to provide women and marginalized groups with an opportunity to express their opinions.
Working conditions	No risk identified	Occupational Health and Safety will be prioritized at every stage of the project. Employees will receive regular training to ensure their health and safety in the workplace. Measures will also be implemented to safeguard the health of the community near the project area. The project is committed to complying with national Occupational Health and Safety Standards, national labour standards, IFC PS2 and International Labour Organization Labor Standards, following all guidelines and standards.
Safety of women and girls	No risk identified	The project promotes safe and secure environments for all stakeholders including women and girls. The proposed project is a renewable energy grid connected power project and therefore it's not gender sensitive and does not adversely impact the safety of women and girls
Safety of minority and marginalized groups, including children	No risk identified	The project promotes safe and secure environments for all stakeholders including minority, marginalized groups and children

Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	No risk identified	The project is a solar energy project, hence will not result in release of pollutants to the environment. The project will have a temporary impact on the environment only during the construction phase due to soil disruption, particulate dust generation and noise associated with vehicles, respective measures have been taken to minimize the impact. All waste from land clearing, building activities, and accommodation were collected, stored and disposed of safely in a manner that prevents any release of leachate. The project ensures proper handling, storage, and disposal of redundant/broken panels through designated broken PV storage room at the project site. Disposal will comply with Sub-decree No. 36 on Solid Waste Management (MOE, 1999), utilizing licensed vendors for collection, transportation, and recycling.
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2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Discrimination and sexual harassment	The project is committed to ensuring a work environment free from discrimination and sexual harassment. The project developer has an Anti-sexual harassment policy in place to prohibit any form of discrimination or harassment based on factors such as race, gender, age, religion, or any other protected characteristic. All employees are made aware of these policies through training programs, and clear reporting mechanisms are in place to address any concerns or incidents promptly
Management experience	Prime Road Group, the project developer, brings extensive experience to the table with a track record of successfully managing various regional and international solar development

	projects. This includes 9 projects in Thailand, 5 in Japan, and 1 in Taiwan. These projects highlight the management team's proficiency in promoting renewable energy development and contributing to the sustainable development of local communities.
Gender equity in labor and work	The project developer has a Non-Discrimination and Equal Opportunity Policy in place, promoting equal opportunities, non-discrimination, and fair treatment of workers. The project is committed to eliminating gender-based discrimination and ensuring that everyone, regardless of gender, has equal access to employment, training, and advancement opportunities. Moreover, the project follows transparent and non-discriminatory hiring practices and has set up a fair compensation structure based on principles of pay equity, ensuring that individuals receive equal pay for equal work.
Human trafficking, forced labor, and child labor	Prime Road Alternative (Cambodia) Co., Ltd and its sub-contractors strictly adhere to all national and international laws related to child labor, forced labor, and human trafficking. The project unequivocally affirms that it does not and will not use victims of human trafficking, forced labor, or child labor in any aspect of its operations. This commitment aligns with the project's dedication to upholding ethical and legal standards in its workforce practices.

2.3.2 Human Rights

The project avoids any activities that could violate the economic, social, and cultural well-being of local communities. Consequently, the project developer and the project itself commit to respecting internationally proclaimed human rights. They will not be complicit in any form of violence or human rights abuses, as defined in the Universal Declaration of Human Rights. This commitment reflects the project's dedication to upholding ethical standards and safeguarding the well-being of the communities involved.

2.3.3 Indigenous Peoples and Cultural Heritage

The Project will not directly or indirectly affect the dignity, human rights, livelihood systems, or culture of Indigenous Peoples (IPs), neither will it affect the territories or natural or cultural resources that IPs own, use, occupy, or claim as their ancestral domain.

2.3.4 Property Rights

Rights to territories and resources	This project is a component of Electricite du Cambodge's (EDC) National Solar Park (NSP) initiative. The project proponent has secured a 97-hectare lease for a 60 MW photovoltaic (PV) plant from EDC. EDC acquired the land through commercial negotiations, following a willing buyer-willing seller approach. As a result, the project site does not involve any territories or resources to which the stakeholders have customary rights or access.
Respect for property rights	Not applicable as the project does not involve any territories or resources to which the stakeholders have customary rights or access.

2.3.5 Benefit Sharing

As described in section 2.3.4 above, the project developer has leased the land from EDC, which, in turn, acquired the land through commercial negotiations and holds all rights to the land on which the project has been developed. Further, the project does not involve any territories to which the stakeholders have customary rights and therefore benefit sharing is not applicable and hence no benefit sharing plan has been designed.

Process used to design the benefit sharing plan	NA
Summary of the benefit sharing plan	NA
Approval and dissemination of benefit sharing plan	NA
Benefit sharing during the monitoring period	NA

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	No risk identified	ESIA has confirmed that the projected area is not located in sensitive ecological zones,

		biodiversity conservation areas, and there are no rare and valuable plant and animal species.
Soil degradation and soil erosion	Risk of soil erosion during construction	The project might lead to soil erosion during the construction phase. However, mitigation measures will be implemented, such as recovering topsoil stripped during clearing and construction. Land clearing will be confined to the specific project site needed for development, minimizing the impact. Activities involving chemicals or hydrocarbons, including the storage of broken or redundant solar panels, will be limited to designated areas with proper bunding, a concrete or compacted earth base, and complete containment by drainage controls. Furthermore, soil quality monitoring and testing was carried out for six months during the construction period.
Water consumption and stress	No risk identified	The project will not contaminate or impede the natural flow of any surface or sub-surface water sources. Additionally, the project incorporates a water harvesting procedure to collect stormwater and drained into water retention ponds. Furthermore, ground and surface water quality tests were conducted both during the construction and operation phase.
Usage of fertilizers	No risk identified	The project harnesses solar energy to generate electricity and therefore, will not involve application of fertilizers, pesticides or herbicides

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes

☒ No

Species and habitat	NA
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2.4.2 Introduction of species

This is not a AFOLU project and doesn't involve introduction of any species into the project area, hence this section is Not Applicable

Species introduced	Classification	Justification for use	Adverse effects and mitigation
NA	NA	NA	NA

Existing invasive species	Mitigation measures to prevent spread or continued existence of invasive species
NA	NA

2.4.3 Ecosystem conversion

The project activity has not resulted in conversion or clearing of the existing ecosystem

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	21.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	7.0
Tool	TOOL01	Tool for the demonstration and assessment of additionality	7.0.0
Tool	TOOL27	Investment analysis	13.0
Tool	TOOL24	Common Practice	03.1

3.2 Applicability of Methodology

This project is a large-scale grid-connected Greenfield renewable energy power plant. Large-scale Approved Consolidated Methodology ACM0002: Grid-connected electricity generation from renewable sources -- Version 21.0⁹ is applicable for the proposed project as per the scope defined in the methodology. A detailed discussion regarding applicability of ACM0002 methodology, Version 21.0 given in table below.

Methodology ID	Applicability condition	Justification of compliance
ACM0002	4. This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	The project activity is a Greenfield solar power plant and hence the project meets the applicability condition (a).
	5. In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s)	Not Applicable as the project doesn't involve integration of BESS

⁹ <https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL01OU2RYC6N3HASIXV7K84QBG9>

	without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	
	6. The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such	The project involves installation of a solar power plant and is applicable under the condition (a)

	cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	
	7. In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated	Not applicable as this project is a solar power plant

	project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
	8. In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	It's not applicable as the proposed project is a solar power project
	9. The methodology is not applicable to:	The proposed project is neither fuel switch nor

	(a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	biomass plant as it is a solar power plant. Hence, the methodology is applicable for the project activity.
	10. In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”	This condition is not applicable as the project doesn’t involve any retrofits, capacity addition rehabilitations or replacements in the existing plant/unit.
In addition to satisfying the applicability criteria of the methodology ACM0002, the project also meets the applicability conditions of the tools applied as below:		
TOOL01	9. The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Applicable The project participant is not using any new methodology, this tool has been applied to demonstrate additionality.
	10. Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	Applicable as the methodology used by the project requires the use of this tool.
TOOL07	3. This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity	Applicable as this project is a greenfield solar power project which supplies renewable

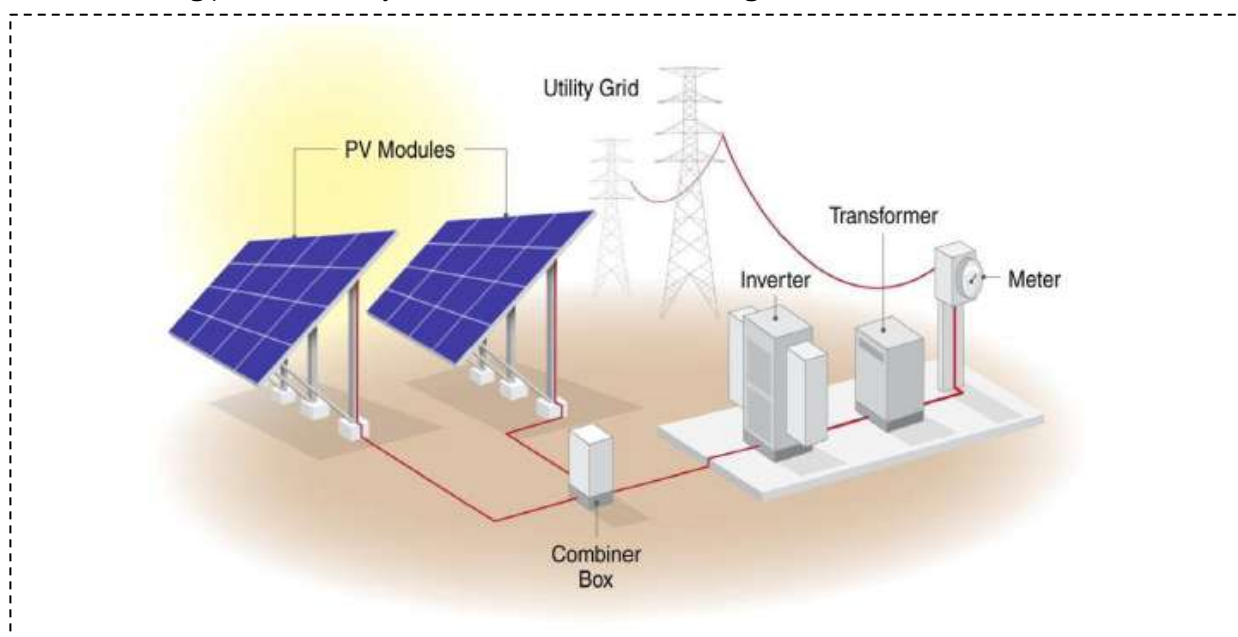
	that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	energy to the grid and this tool is used to calculate the OM, BM & CM of the project activity.
	4. Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Combined Margin Emission Factor cannot be calculated under this tool as the data required to calculate Operating Margin (OM) is unavailable as Specific fuel consumption data for the Cambodia’s national power grid over the past five years is not publicly accessible, and the default grid emission factors published by Cambodia’s Ministry of Environment are based on data from 2010, 2011, and 2012. A conservative default Combined Margin Grid Emission Factor of 0.5878 tCO₂/MWh as available from the most recent version of the Harmonized IFI Default Grid Factors 2021 (Version 3.2) will be used instead to quantify the GHG emission reductions occurring due to this project.
	5. In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Not Applicable, the project is located in Cambodia, a non-Annex I country.

	6. Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	Not Applicable as the project is a solar power plant.
TOOL27	2. This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	Applicable as this project activity applies the methodological tool, “Tool for the demonstration and assessment of additionality.”
	3. In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied methodology, ACM0002 Version 21.0 requires the use of this tool to demonstrate investment analysis of this project.
TOOL24	3. This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	Applicable as this project activity applies the methodological tool, “Tool for the demonstration and assessment of additionality.”
	4. In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different	The applied methodology, ACM0002 Version 21.0 requires the use of this tool to demonstrate the common practice of this project.

from those described in this methodological tool, the requirements contained in the methodology shall prevail.

3.3 Project Boundary

According to the methodology, the spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the project power plant is connected to. Hence, the project boundary includes the project site where the power plant has been installed, associated power evacuation infrastructure, energy metering points, switch yards and the connected EDC grid.



Source: National Renewable Energy Laboratory (NREL)¹⁰

Figure 2: Project Boundary Diagram

¹⁰ <https://www.nrel.gov/docs/fy16osti/67122.pdf>

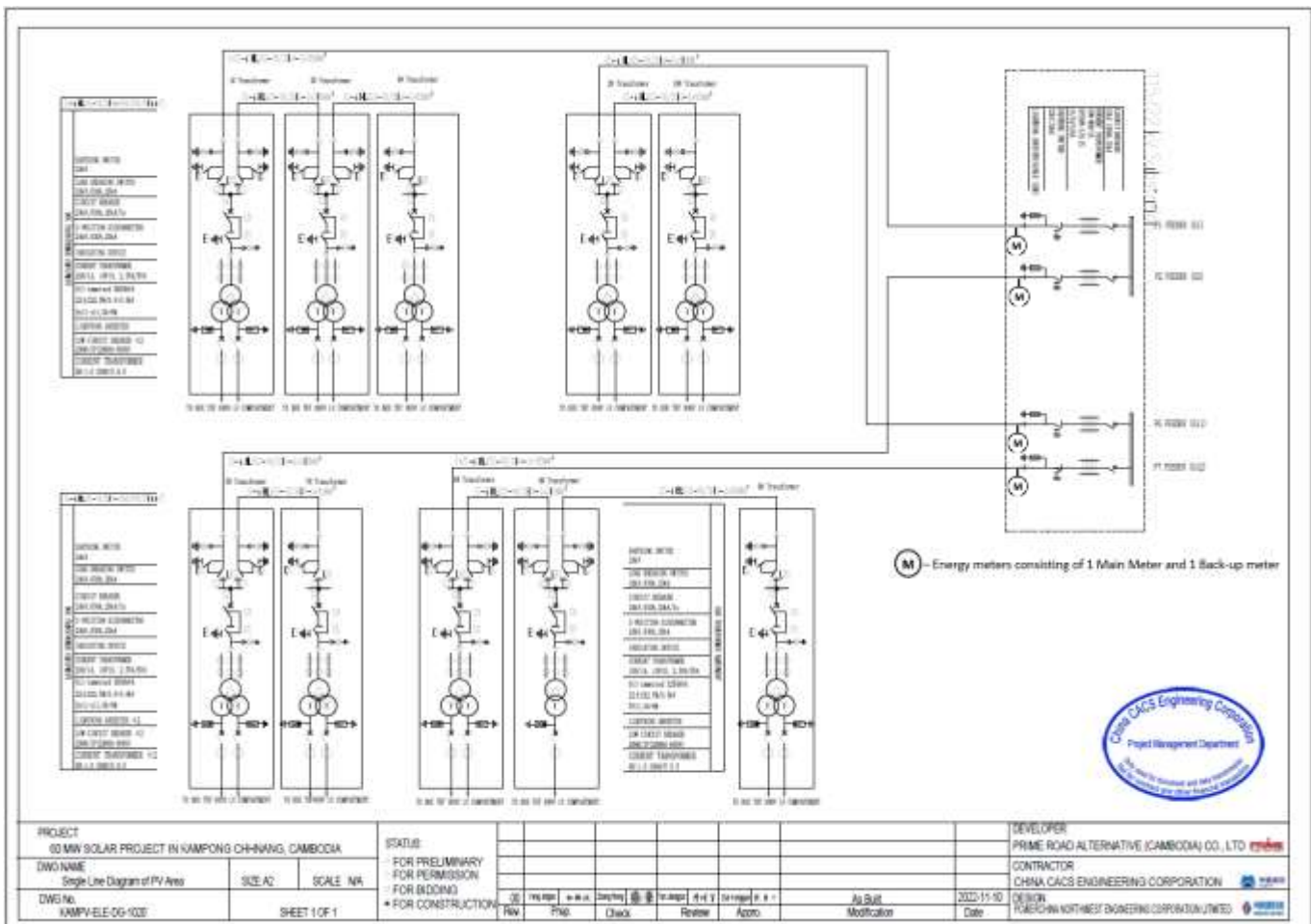


Figure 3: Single Line Diagram of PV Area

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions:

Source		Gas	Included?	Justification/Explanation
Baseline	Source 1: CO ₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO ₂	Yes	Main source of emission in the baseline
		CH ₄	No	Minor source of emission
		N ₂ O	No	Minor source of emission
		Other	No	Minor source of emission
Project	Greenfield Solar Power Project activity	CO ₂	No	As per ACM0002, for renewable energy power generation project emissions are PE _Y = 0
		CH ₄	No	
		N ₂ O	No	
		Other	No	

3.4 Baseline Scenario

According to ACM0002 - Version 21.0, if the project involves installing a Greenfield power plant, the baseline scenario is the electricity delivered to the grid by the project activity. This electricity would have otherwise been generated by the operation of grid-connected power plants and the

addition of new generation sources, as outlined in the combined margin (CM) calculations described in "TOOL07: to calculate the emission factor for an electricity system¹¹."

The combined margin ($EF_{grid,CM,y}$) represents a weighted average of two emission factors related to the electricity system: the operating margin (OM) and build margin (BM). The calculation of the combined margin must be based on data from an official source, where available, and made publicly accessible. Due to the non availability of the latest grid emission factor approved by the Designated National Authority (DNA) of Cambodia, a Combined Margin Grid Emission Factor of 0.5878 tCO₂/MWh will be applied for GHG reduction quantification of this project activity which is based on the most recent version of the Harmonized IFI Default Grid Factors 2021 (Version 3.2). Details of the same has been outlined below in section 3.6.

The combined margin of the grid used for the project activity is as follows:

Parameter	Value (tCO ₂ /MWh)	Description	Source
$EF_{grid,CM,y}$	0.5878	Combined margin CO ₂ emission factor for the project electricity system in year y	CM is based on the IFI Technical Working Group (IFI TWG) dataset containing Default Emission Factors for electricity grid, and is sourced from the most recent version of the Harmonized IFI Default Grid Factors 2021 (Version 3.2)

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☐ No

Not Applicable since the project activity is not mandated by any law or regulations and is entirely voluntary in nature.

¹¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

3.5.2 Additionality Methods

As per ACM0002 – Version 21.0, the additionality of a project activity shall be demonstrated and assessed using the “TOOL01 - Tool for the demonstration and assessment of additionality¹².”

The tool provides a step-wise approach to demonstrate additionality as described below:

Step 0: Demonstration whether the proposed project activity is the first-of its kind

This step is not applied to the project activity since it is not first-of-its-kind, hence the additionality of the project will be demonstrated in next steps below.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity

Alternatives available to the project participants or similar project developers that provide outputs or services comparable with the proposed CDM project activity. These alternatives are as below:

Alternative (a) The proposed project activity undertaken without being registered as a VCS project activity.

Alternative (c) Continuation of the current situation (no project activity or other alternatives undertaken).

Outcome of Step 1a: Alternative (a) & (c) are realistic and credible alternative scenario(s) to the project activity

Sub-step 1b: Consistency with mandatory laws and regulations

Under Cambodia's Electricity Law, there are no legal restrictions that prevent the implementation of renewable energy projects, ensuring that alternative (a) is fully in line with the country's laws and regulations.

Alternative (b), which reflects the continuation of the baseline scenario where electricity supplied to the EDC grid by the project would have otherwise been generated by grid-connected power plants or additional generation sources, also complies with Cambodian regulations. The Electricity Law of the Kingdom of Cambodia does not prohibit the grid from delivering the same amount of electricity annually.

Outcome of Step 1b: Both alternatives (a) and (b) realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations

Step 2: Investment analysis

As per paragraph 29 of "Tool for the demonstration and assessment of additionality" version 7.0.0, it is determined that the proposed project activity is not an economically or financially

¹² <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

feasible option. Referring to TOOL27 - Investment analysis, version 13.0 (EB 120 annex 3)¹³ for the investment analysis, the financial analysis of the project considers three options, as outlined in "Tool for the demonstration and assessment of additionality" (version 07.0.0):

Sub-step 2a: Determine appropriate analysis method:

Option I: Simple Cost Analysis

Option II: Investment Comparison Analysis

Option III: Benchmark Analysis

Since the project generates revenues from the sale of electricity, Option I is not applicable. Option II is also not applicable because there is no comparable investment alternative available to the project participant. Therefore, the most appropriate financial analysis method is Option III: the benchmark analysis, comparing returns on investment in the project activity to benchmark returns available to any investors in the country.

Sub-step 2b: Option III. Apply benchmark analysis

The project proponents have taken into account the Post-Tax Equity IRR for investment analysis during decision-making. The decision-making date is the EPC signing date which is 13th August 2021. Since the EPC signing was the first binding step that led to the finalization of the land lease and financing—both essential for the PPA's effectiveness—the EPC contract signing date is considered the investment decision date. The project proponent focuses on the returns generated on the portion of investment costs financed by them in the form of equity. According to Para 15 of TOOL 27: Investment analysis, Version 13.0, Required/expected returns on equity are suitable benchmarks for an equity IRR. Hence, the Expected return on equity is considered an appropriate benchmark. Consequently, the post-tax Equity IRR is regarded as the relevant financial indicator for Investment Analysis.

Default Value Benchmark

According to paragraph 19 of TOOL 27: Investment analysis, Version 13.0, the cost of equity is determined by choosing values provided in the Appendix. Default values for the cost of equity (expected return on equity) are presented below:

The Required return on equity (benchmark) was computed in the following manner:

$$\text{Nominal Benchmark} = \{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$$

Where:

– Default value for Real Benchmark is the default value of expected return on equity in real terms for Energy Industries (Group 1) in Cambodia as provided in the Appendix

– Average forecasted inflation rate for the host country published by the IMF for the next five years after the start of the project activity

Benchmark estimation

The Cost of Equity has been determined using the "Methodological tool: Investment analysis" available at the time of decision-making, as well as the latest available value. As a conservative

¹³ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v13.pdf>

approach, the minimum value of the benchmark has been considered, calculated using these two approaches.

Default Value at the time of investment decision:

Appendix, Investment Analysis, EB105, Annex 06¹⁴ specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Cambodia = 15.24%

Default Value as per latest version of Investment Analysis Tool version 13.0:

Table under EB 120 annex 3 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Cambodia = 14.15%

Thus, minimum Default Value considered for calculation of Benchmark = 14.15%

As per Para 16 of TOOL 27: Investment analysis, Version 13.0, the average forecasted inflation rate for Cambodia published by the IMF (International Monetary Fund, World Economic Outlook) for the next five years after the start of the project activity has been used.

The five-year inflation forecast for Cambodia as published by IMF in its World Economic Outlook Database and the corresponding benchmark value is 2.95%

Average forecasted inflation rate for 5 years	Source
2.95%	IMF, World Economic Outlook Database ¹⁵

The nominal benchmark has been calculated as; $\{(1+14.15) * (1+2.95)\} - 1 = 17.52\%$

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III):

Input values used in the investment analysis:

Particulars	Value	Unit	Source/Remarks
Capacity of the project	60	MW	Technical Due Diligence Report
Annual Net generation	141719	MWh	Technical Due Diligence Report
Project cost	41224.142	k USD	Technical Due Diligence Report
Debt	70	%	Technical Due Diligence Report
Equity	30	%	Technical Due Diligence Report
Debt	28856.90	k USD	Calculated
Equity	12367.24	k USD	Calculated
Interest rate	7%	%	Technical Due Diligence Report
Debt Repayment tenure	17	years	Technical Due Diligence Report
O&M Cost	807.039	k USD	Technical Due Diligence Report
Escalation in expense	0.03		Technical Due Diligence Report

¹⁴ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v10.0.pdf>

¹⁵ IMF, World Economic Outlook Database

1st Year tariff	0.03877	USD/kWh	Technical Due Diligence Report
Lifetime of the plant	20	Years	Technical Due Diligence Report
SLM Depreciation Rate (Book)	5%	%	Calculated
Tax holiday	9	Years	Technical Due Diligence Report
Income tax rate from 10th year onwards	20%		Technical Due Diligence Report
Annual Degradation Factor	0.45%		Technical Due Diligence Report

Considering the input values, Equity IRR is calculated as 1.31% against the benchmark of 17.52%. Hence, the project activity cannot be considered as financially attractive as the equity IRR for the project activity is less than the Benchmark.

Sub-step 2d: Sensitivity Analysis

In line with paragraphs 27 and 28 of TOOL 27: Investment analysis, Version 13.0, the following factors have been subjected to sensitivity analysis: 1. Tariff, 2. Generation, 3. Project Cost, and 4. O&M Cost. The rationale for sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the one presented, providing a cross-check on the suitability of the assumptions used in the development of the investment analysis.

The results of the sensitivity analysis are as follows:

Variation %	-10%	Normal	10%	Variation required to reach benchmark	Value required to reach benchmark	
Tariff	-1.23%	1.31%	3.68%	76.60%	0.068	USD/kWh
Generation	-1.23%	1.31%	3.68%	76.60%	250276	MWh
Project Cost	3.44%	1.31%	-0.51%	-48.00%	21436.55	k USD
O&M Cost	1.79%	1.31%	0.82%	NA*	NA*	k USD

*Even with the zero O&M cost the IRR will not reach benchmark. Hence no value is provided

The analysis reveals that even with a variation of +10% and -10% in project cost, O&M cost, generation and tariff rate, Equity IRR is significantly lower than the benchmark. As evident from the results above, the project remains additional even under the most favorable conditions.

Parameters	Probability of breaching the benchmark
Tariff	The tariff considered for the sensitivity analysis is in line with the signed PPA and as per the +10% to -10% sensitivity analysis the calculated IRR is exceeding the benchmark when the tariff changes by 76.60% which unlikely throughout the PPA of 20 years.
Generation	In the first year of operation the project has generated 138,779 MWh of electricity which is in line with the maximum power generation in the first year of operation as per the energy yield assessment done by the technical advisor AFRY (Thailand). Hence, it is highly unlikely that the generation will breach the benchmark value.
Project Cost	The project is already commissioned and any reduction in project cost is impossible.

O&M Cost	The sensitivity analysis reveals that O&M will breach the benchmark at a negative value and is a hypothetical case. Hence, the O&M costs associated with the project has no impact on the financial returns of the equity.
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Outcome of Step 2:

This confirms that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark Equity IRR). Thus, it can be easily concluded that the project activity is additional and is not a business-as-usual scenario.

Step 3: Barrier analysis

Barrier analysis has not been applied

Step 4: Common practice analysis

Stepwise approach for common practice analysis has been carried out as per Methodological tool “Common Practice”, version 03.1 EB84, Annex 7¹⁶:

Definitions as per methodological tool	Justification
Step 1: calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.	+50% of project's installed capacity: 90 MW Installed capacity of the project activity: 60 MW -50% of project's installed capacity: 30 MW
Step 2: identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions: (a) The projects are located in the applicable geographical area; (b) The projects apply the same measure as the proposed project activity; (c) The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity; (d) The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications	(a)The project is located in Kampong Chhnang province, Cambodia (b)The proposed project activity uses renewable energy to generate power. Therefore, all renewable energy projects located in Cambodia are considered. (c)The energy source used by the project activity is Solar. Hence, only solar energy projects have been considered. (d) The project activity produces electricity; therefore, all solar power plants are considered.

¹⁶ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf>

areas (e.g. clinker) as the proposed project plant; (e) The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1; (f) The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.	(e) The capacity range of the projects is within the applicable capacity range from 30 to 90 MW (f) As per CDM, for a project activity, the date on which the project participants commit to make expenditures for construction and where a contract is signed for such expenditures, the date on which the contract is signed is considered the start date and as CPA is being carried out as per the CDM tool, the EPC contract signing date i.e., 13th August 2021 is being considered as the project start date of proposed project activity. As Kyoto Protocol was ratified by Cambodia on 4th July 2002, solar projects which had started commercial operation between 4th July 2002, 13th August 2021 have been identified. Numbers of Similar projects (CDM and non-CDM) identified based on available published sources^{17,18}, which fulfils the above-mentioned conditions are 5 i.e. $N_{\text{solar}} = 5$
Step 3: within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number N_{all}	Within the 5 projects identified in Step 2, 2 projects are registered under the International Renewable Energy Certificates (I-RECs) mechanism¹⁹. Therefore $N_{\text{all}} = 3$
Step 4: within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff}	The identified projects in Step 3 do not apply technologies other than solar. Hence, $N_{\text{diff}} = 0$

¹⁷ <https://cleanenergycambodia.org/cleanenergymap>

¹⁸ <https://eac.gov.kh/site/annualreport>

¹⁹ <https://evident.app/IREC/device-register/table>

Step 5: calculate factor $F=1-N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.	$F=1-N_{diff}/N_{all}$ $F = 1 - 0/3 = 1 - 0$ Hence, F is 1
As per paragraph 18, The proposed project activity is a “common practice” within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all}-N_{diff}$ is greater than 3.	From the above, $F = 1$ which is greater than 0.2 $N_{all}-N_{diff} = 3$ which is equal to 3. Hence, the project activity is not a “common practice” in the applicable geographical area.

Outcome of Step 4:

The above explanation/justification clearly shows that the proposed project activity is unlikely to be financially attractive as well as not a common practice in the host country; hence the project activity is additional.

The major milestones of the project are as below:

Date	Milestone
24 October 2019	Tender Award
30 June 2020	PPA between EDC & Prime
13 August 2021	EPC contract signed
11 November 2022	Project was successfully energized and synchronized

3.6 Methodology Deviations

According to TOOL7, "Tool to calculate the emission factor for an electricity system" (Version 07.0, Section 6.6), if data required to determine the Operating Margin (OM) is unavailable, it cannot be used to calculate the Combined Margin (CM) emission factor. Specific fuel consumption data for the EDC grid over the past five years is not publicly accessible, and the default grid emission factors published by Cambodia's Ministry of Environment are based on data from 2010, 2011, and 2012²⁰.

The IFI Technical Working Group (IFI TWG) on GHG accounting maintains a dataset containing the Default Emission Factors (DEF) for electricity grids, including interconnected grids within a country. These DEFs apply to national electricity generation and currently do not account for

²⁰https://www.iges.or.jp/en/publication_documents/pub/data/en/5097/GEF-Cambodia_2010-2012.pdf

cross-border interconnections. As outlined in "The Methodological Approach for the Common Default Grid Emission Factor Dataset" (Version 01.1)²¹, the common dataset uses a Combined Margin (CM) approach, comprising an Operating Margin (OM) and Build Margin (BM), consistent with Clean Development Mechanism (CDM) guidelines.

The most recent version of the Harmonized IFI Default Grid Factors 2021 (Version 3.2)²² provides a Combined Margin Grid Emission Factor of 0.8743 tCO₂/MWh for intermittent energy sources like solar and wind, and 0.5878 tCO₂/MWh for firm energy sources such as hydro and geothermal. In this project, a conservative approach will use a combined margin of 0.5878 tCO₂/MWh to calculate emission reductions. This deviation will not affect the quantification of GHG emission reductions resulting from this project activity.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The current status of project implementation includes the following key points:

Operation during Monitoring Period:

The project commenced its operation on November 11, 2022, and has been consistently operational throughout the monitoring period. Monthly net electricity supplied to the grid, crucial for emission reduction calculation, is extracted from the Joint Meter Reading (JMR) reports. No significant issues affecting emission reductions or removals have been identified during this period.

Changes to Project Proponent or Other Entities:

There have been no alterations to the project proponent or other essential entities associated with the project. The original project proponent continues to lead and oversee the project's implementation.

²¹ https://unfccc.int/sites/default/files/resource/IFITWG_Methodological_approach_to_common_dataset.pdf

²² <https://unfccc.int/climate-action/sectoral-engagement/ifis-harmonization-of-standards-for-ghg-accounting/ifi-twag-list-of-methodologies>

5 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology ACM0002 – Ver21.0, assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants.

The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (t CO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y is calculated using TOOL07 – Tool to calculate the emission factor for an electricity system, Version 07.0 (t CO₂/MWh)

Baseline emission factor is calculated as combined margin, consisting of a combination of operating margin and build margin factors according to the procedures prescribed in the latest “tool for calculating the emission factor for an electricity system”, version 07.0.

However, due to the unavailability of specific fuel consumption data for Cambodia’s national power grid for the past five years, along with the lack of an updated grid emission factor approved by the Designated National Authority (DNA) of Cambodia, a Combined Margin Grid Emission Factor of 0.5878 tCO₂/MWh will be utilized for the GHG reduction calculations for this project and will remain constant throughout the crediting period. This factor is based on the most recent Harmonized IFI Default Grid Factors 2021 (Version 3.2).

5.2 Project Emissions

As per the ACM0002 ver-21.0, Project Emission for most renewable energy power generation project activities, $PE_y = 0$. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} + PE_{BESS,y}$$

Where:

PE_y = Project emissions in year y (t CO₂e/yr)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (t CO₂/yr)

$PE_{GP,y}$ = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (t CO₂e/yr)

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants in year y (t CO₂e/yr)

$PE_{BESS,y}$ = Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (t CO₂e/yr)

The project activity involves the generation of electricity from the installation of solar projects. Hence, as per ACM0002, ver-21.0, there is no project emission for solar projects. Therefore, project emissions are zero.

5.3 Leakage Emissions

The project activity does not account for any leakage emissions. In the context of electric sector projects, potential emissions contributing to leakage, such as those from power plant construction and upstream fossil fuel use (including extraction, processing, and transport), are deemed negligible due to their limited scale and are therefore not considered in the assessment.

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

As per, ACM0002 ver-21.0, the ex-ante emission reductions (ER_y) for the project activity are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (tCO₂e/yr)

BE_y = Baseline emissions in year y (tCO₂e/yr)

PE_y = Project emissions in year y (tCO₂e/yr)

Since $PE_y = 0$

$$ER_y = BE_y$$

In first crediting period, the total estimated emission reduction of the project is 573,078 tCO₂e, with average annual emission reductions of 81,868 tCO₂e.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
11-Nov-2022 to 31-Dec-2022	11,639	0	0	11,639	NA	11,639
01-Jan-2023 to 31-Dec-2023	82,928	0	0	82,928	NA	82,928
01-Jan-2024 to 31-Dec-2024	82,555	0	0	82,555	NA	82,555
01-Jan-2025 to 31-Dec-2025	82,183	0	0	82,183	NA	82,183
01-Jan-2026 to 31-Dec-2026	81,813	0	0	81,813	NA	81,813
01-Jan-2027 to 31-Dec-2027	81,445	0	0	81,445	NA	81,445
01-Jan-2028 to 31-Dec-2028	81,079	0	0	81,079	NA	81,079
01-Jan-2029 to 10-Nov-2029	69,436	0	0	69,436	NA	69,436

Total	573,078	0	0	573,078	NA	573,078
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6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined Margin CO ₂ emission factor in year y
Source of data	Harmonized IFI Default Grid Factors 2021 (Version 3.2) ²³
Value applied:	0.5878
Justification of choice of data or description of measurement methods and procedures applied	Due to the unavailability of specific fuel consumption data for Cambodia's national power grid for the past five years, along with the lack of an updated grid emission factor approved by the Designated National Authority (DNA) of Cambodia, a Combined Margin Grid Emission Factor of 0.5878 tCO ₂ /MWh will be utilized for the GHG reduction calculations for this project. This factor is based on the most recent Harmonized IFI Default Grid Factors 2021 (Version 3.2). Details of the same has been provided in above in section 3.6.
Purpose of data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period

²³ <https://unfccc.int/climate-action/sectoral-engagement/ifi-harmonization-of-standards-for-ghg-accounting/ifi-twag-list-of-methodologies>

6.2 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$																			
Data unit	MWh/year																			
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y in MWh																			
Source of data	Monthly joint meter reading reports																			
Description of measurement methods and procedures applied	Measurements are undertaken using energy meters. The monthly net electricity supplied to the grid by the project activity is determined by subtracting the final value of exports from imports. This same value will be utilized for emission reduction calculations. The metering equipment is supplied and installed by EDC and all the billing meters used in the incoming feeders have an accuracy class of 0.5s.																			
Frequency of monitoring/recording	Continuous monitoring & monthly recording																			
Value applied:	141,719																			
Monitoring equipment	Electricity meters, the details of which are as below: <table><tr><th>Feeder</th><th>Main Meter Number</th><th>Backup Meter Number</th><th>Accuracy</th><th>Make</th></tr><tr><td>1</td><td>55548513</td><td>55548515</td><td rowspan="4">0.5s</td><td rowspan="4">Landis+Gyr E650</td></tr><tr><td>2</td><td>55548514</td><td>55580597</td></tr><tr><td>6</td><td>55548500</td><td>55580600</td></tr><tr><td>7</td><td>55548498</td><td>55548501</td></tr></table>	Feeder	Main Meter Number	Backup Meter Number	Accuracy	Make	1	55548513	55548515	0.5s	Landis+Gyr E650	2	55548514	55580597	6	55548500	55580600	7	55548498	55548501
Feeder	Main Meter Number	Backup Meter Number	Accuracy	Make																
1	55548513	55548515	0.5s	Landis+Gyr E650																
2	55548514	55580597																		
6	55548500	55580600																		
7	55548498	55548501																		
QA/QC procedures applied	The State Utility EDC,has approved, tested, and sealed the meters, keeping them in its custody. The meters undergo calibration once every year as per the calibration certificate. The monthly electricity supplied or exported by the project activity, as reported in the Joint Meter Reading (JMR) report, is cross-checked with the monthly sales invoices. In case of meter calibration absence or delay, appropriate guidelines will be applied to confirm the conservatism of metering. The metering arrangement, accuracy class of meters, and calibration frequency are under the control of the EDC, and																			

	the project proponent (PP) has no authority over them. The PP receives the value of net electricity supplied to the grid, which serves as the monitoring parameter, while billing is raised based on substation meters.
Purpose of data	Calculation of baseline emissions
Calculation method	Net electricity supplied to the grid by the project plant in a given month = Export(kWh)– Import (kWh).
Comments	Data will be archived in both paper and electronic forms for two years after the end of the crediting period or the last issuance of Verified Emission Reductions (VERs) for this project activity, whichever occurs later.

6.3 Monitoring Plan

The monitoring plan aims to make sure the project accurately measures emission reductions during its crediting period. The project owner, with support from EDC, will carry out this plan. It ensures that actual, measurable, and long-term reductions in greenhouse gas (GHG) emissions are well-tracked and reported, playing a vital role in determining the project's overall emission reductions. The project owner will put this monitoring plan into action during the project's operation, and the details of the same as below:

Methods used for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters

Net electrical energy generated by the facility is measured and metered in kWh at the Interconnection Point at the EDC substation where the energy meters are supplied and installed by the EDC. The main meters are used to measure the Net Electrical Output, except during any period when such meters are out of service as a result of maintenance, repairs or testing, in which event, the back-up meters are be used to measure the Net Electrical Output. The project proponent will be allowed to remotely carry out readings of the billing meters located at EDC's premises on an hourly basis. EDC shall be responsible for testing and calibrating the Metering System for accuracy. These meters are calibrated on a yearly basis as per the calibration certification and has an accuracy class of 0.5s.

Meters will be read remotely by both Parties and EDC shall record the show values on the active energy meters and reactive energy meters on the first day of a month. The Authorized Representative of the Company is entitled to be present to supervise the process in order to verify the quantity of electricity delivered by the project to EDC in each of the monthly billing cycle.

The organizational structure, responsibilities and competencies of the personnel that carried out monitoring activities

Meters are jointly read by both the EDC official at the substation where the meters are located along with the representative of the project proponent who will be present in order to verify the

quantity of electricity delivered by the project to EDC on the first day of a month. All data will be signed by the reader and checker and will be photographed or printed. Further, the finance department of the project proponent will cross check the JMR reading with sales invoices and make a regular summary of the data and maintains an archive of the same.

QA/QC

All four feeders have a main and a back-up meter, in case of main meter failures back-up meters are used to measure the net electricity output. The meters are all under the jurisdiction of EDC and is responsible for undertaking annual calibration tests and in any case the metering equipment's are found to be inaccurate, EDC is responsible to carry out repairs and/or replacement.

Procedures for handling non-conformances with the validated monitoring plan

In case of non-conformances such as malfunction of the metering equipment, EDC will provide technical assistance and is responsible to carry out repair or replacement of the equipment. Any emission reductions quantified during such period will follow a conservative approach.

Data storage and archiving

All the data items monitored will be kept for two years after the end of crediting period or till the last issuance of VERs for this project activity whichever occurs later. The monitored data will be presented for future verifications.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh/y
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y in MWh.
Value applied:	Measured: 156,367 After application of error: 156,242
Comments	The calibration details of the energy meters are given below:

Feeder	Main Meter Number	Backup Meter Number	Commissioning date/Calibration date	Validity
1	55548513	55548515	11/11/2022	10/11/2023
2	55548514	55580597		
6	55548500	55580600		
7	55548498	55548501		
1	55548513	55548515	20/03/2024	19/03/2025
2	55548514	55580597	21/03/2024	20/03/2025
6	55548500	55580600	22/03/2024	21/03/2025
7	55548498	55548501	22/03/2024	21/03/2025

From the above-mentioned calibration data, it is found that the calibration is not valid for the period from 10/11/2023 to 31/12/2023 during the monitoring period. Hence, the maximum possible error of 0.5% is applied for the November & December 2023 data as below:

Export = Measured value * (1-0.5%)

Import = Measured value *(1+0.5%)

Data will be archived in both paper and electronic forms for two years after the end of the crediting period or the last issuance of Verified Emission Reductions (VERs) for this project activity, whichever occurs later.

7.2 Baseline Emissions

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y as per the most recent Harmonized IFI Default Grid Factors 2021 (Version 3.2).(tCO₂/MWh)

Monitoring Period	$EG_{PJ,y}$ (MWh)	$EF_{grid,CM,y}$ (tCO ₂ /MWh)	BE_y (tCO ₂ /year)
11-Nov-2022 to 31-Dec-2022	17,463	0.5878	10,264
01-Jan-2023 to 31-Dec-2023	138,779	0.5878	81,575
Total	156,242	0.5878	91,839

7.3 Project Emissions

Not Applicable, since emissions from the project activity is zero.

7.4 Leakage Emissions

Not Applicable, since emissions from the project activity is zero.

7.5 GHG Emission Reductions and Carbon Dioxide Removals

As per, ACM0002 ver-21.0, the emission reductions (ER_y) for the project activity are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (tCO_{2e}/yr)

BE_y = Baseline emissions in year y (tCO_{2e}/yr)

PE_y = Project emissions in year y (tCO_{2e}/yr)

Since PE_y = 0

$$ER_y = BE_y$$

Vintage period	Baseline emissions (tCO _{2e})	Project emissions (tCO _{2e})	Leakage emissions (tCO _{2e})	Reduction VCUs (tCO _{2e})	Removal VCUs (tCO _{2e})	Total VCUs (tCO _{2e})
11-Nov-2022 to 31-Dec-2022	10,264	0	0	10,264	NA	10,264
01-Jan-2023 to 31-Dec-2023	81,575	0	0	81,575	NA	81,575
Total	91,839	0	0	91,839	NA	91,839

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
----------------	---------------------------------------	------------------------------	--------------------	--------------------------------

11-Nov-2022 to 31-Dec-2022	11,639	10,264	-11.81%	The achieved emission reductions are lower than the estimated reductions. As the project is a solar project, the electrical power output is directly influenced by solar irradiance. A decrease in this factor results in lower generation, impacting the emission reductions.
01-Jan-2023 to 31-Dec-2023	82,928	81,575	-1.63%	
11-Nov-2022 to 31-Dec-2023	94,567	91,839	-2.88%	
Total	94,567	91,839	-2.88%	

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

Section	Information	Justification
NA	NA	NA

APPENDIX 2: EVIDENCE OF SDG 8



PRIME ROAD ALTERNATIVE (CAMBODIA) CO., LTD
TIN: L001-902000225
No. 20 Street 181, Sangkat Tumnob Tuek, Khan Boeung Keng Kang,
Phnom Penh, Kingdom of Cambodia

TO WHOM IT MAY CONCERN

Prime Road Alternative (Cambodia) Co. Ltd has developed and is operating a 60 MW solar photovoltaic power plant in Kampong Chhnang Province, Cambodia. This project is currently listed and under validation in the Verra Voluntary Carbon Standard (VCS – 4869) under the title, “Kampong Chhnang 60 MW Solar Power Project.”

Throughout its construction and operational phases, the project has created numerous job opportunities, thus demonstrating a tangible and direct contribution to SDG 8 – Decent Work and Economic Growth.

Prime Road Alternative (Cambodia) Co. Ltd hereby explicitly states that during the monitoring period spanning from 11th November 2022 to 31st December 2023, the project has generated employment opportunities as outlined below:

Monitoring Period	Permanent Employees	Temporary Employees	Total Employees
11-Nov-2022 to 31-Dec-2022	Male - 25	Male - 15	41
	Female - 1	Female - 0	
01-Jan-2023 to 31-Dec-2023	Male - 25	Male - 16	42
	Female - 1	Female - 0	

Sincerely,

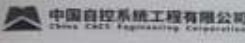
Prime Road Alternative (Cambodia) Co. Ltd



Somprasong Panjalak
Chairman of the Board of Directors
Prime Road Alternative (Cambodia) Co., Ltd

APPENDIX 3: EVIDENCE OF STAKEHOLDER CONSULTATION

Attendance record of the Stakeholder Consultation held on 21st July 2022



中国自控系统工程有限公司
China CACS Engineering Corporation



Attendance Record

Owner	PRAC	Document Number	PRAC-0281072022
EPC Contractor	CACS	Date	21 July 2022
Project Name:	NSP	Project Location	Tpong, KPC

No	Name	Sex	Signature
1	Mr. [Name]	M	[Signature]
2			
3	Mr. [Name]	M	[Signature]
4	Mr. [Name]	M	[Signature]
5	Mr. [Name]	M	[Signature]
6	Mr. [Name]	M	[Signature]
7	Mr. [Name]	M	[Signature]
8	Mr. [Name]	M	[Signature]
9	Mr. [Name]	M	[Signature]
10	Mr. [Name]	M	[Signature]
11	Mr. [Name]	M	[Signature]
12	Mr. [Name]	M	[Signature]
13	Mr. [Name]	M	[Signature]
14	Mr. [Name]	M	[Signature]
15	Mr. [Name]	M	[Signature]
16	Mr. [Name]	M	[Signature]
17	Mr. [Name]	M	[Signature]
18	Mr. [Name]	M	[Signature]
19	Mr. [Name]	M	[Signature]
20	Mr. [Name]	M	[Signature]
21	Mr. [Name]	M	[Signature]
22	Mr. [Name]	M	[Signature]
23	Mr. [Name]	M	[Signature]
24	Mr. [Name]	M	[Signature]
25	Mr. [Name]	M	[Signature]
26	Mr. [Name]	M	[Signature]
27	Mr. [Name]	M	[Signature]
28	Mr. [Name]	M	[Signature]
29	Mr. [Name]	M	[Signature]

Appendix 3: Project Information Banner

Project Manager: Sopheak Kong

HSE Manager: Felix Lan, Rishat Svang

Site Manager: Jeremy Luu



Scanned with CamScanner

Village Level Consultation: Royeas Village, Chean Lerng Commune, Samkymeanchey District,
Kampong Chhnang Province



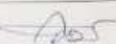
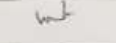
List of Attendees in the village level FPIC (27 September 2020)


ក្រុមហ៊ុន វិនិយោគ និង ដោះស្រាយ បំណុល
Assessment & Solution Co., LTD

Village Consultation in Royas Village
 Chhlean Leony Commune,
 Samaki Banchhay, Kampong
 Chhnang Province
 27/09/2020

បញ្ជីអ្នកមានប្រកបបុគ្គល

ការប្រជុំប្រគល់សិទ្ធិប្រើប្រាស់ដីសម្រាប់គម្រោង
 កាលបរិច្ឆេទ: ២៧ កញ្ញា ២០២០ ទីកន្លែង: ភូមិ រយាតា ឃុំ ឈោង ខេត្ត កំពង់ចាម

ល.រ	ឈ្មោះ	ភេទ	ភូមិ	ទូរស័ព្ទ	ហត្ថលេខា
០១	គ្រី ឈីង	ប	ភូមិ រយាតា	០១១ ៥៨៤៦៥៥	
០២	គ្រី ឈីង	ប	ភូមិ រយាតា	០១០ ២៥៥៨៤៨	
០៣	គ្រី ក	ប	រយាតា	០៩៦ ៤៣៨៤៩១	
០៤	គ្រី ចំណាត	ប	រយាតា	០៩៣ ៦៩៥៤៤៥	
០៥	គ្រី ចំណាត	ប	រយាតា	០៨១ ៦៦៩១៩៣	PR Se
០៦	គ្រី ឈីង	ប	រយាតា		
០៧	គ្រី ឈីង	ប	រយាតា		


ក្រុមហ៊ុន វិនិយោគ និង ដោះស្រាយ បំណុល
Assessment & Solution Co., LTD

ល.រ	ឈ្មោះ	ភេទ	ភូមិ	ទូរស័ព្ទ	ហត្ថលេខា
០៨	គ្រី ឈីង	ប	រយាតា		
០៩	គ្រី ឈីង	ប	រយាតា	០៩៦ ៧១ ៥០៩២២	
១០	គ្រី ឈីង	ប	រយាតា		
១១	គ្រី ឈីង	ប	ភូមិ រយាតា	០៨៥ ២០៦៣០៣	
១២	គ្រី ឈីង	ប	ភូមិ រយាតា	០១៦ ៨៣៤២៣៧	
១៣	គ្រី ឈីង	ប	ភូមិ រយាតា		
១៤	គ្រី ឈីង	ប	ភូមិ រយាតា		
១៥	គ្រី ឈីង	ប	ភូមិ រយាតា		
១៦	គ្រី ឈីង	ប	រយាតា		
១៧	គ្រី ឈីង	ប	រយាតា	០៩៨ ២៣៨៣១៦	



ក្រុមហ៊ុន អ៊ីសេស៊ីយ៉ូន & សំណួរដោះស្រាយ
Assessment & Solution Co., LTD

បេក្ខជនប្រធានប្រតិបត្តិ

Village Consultation
in Trapping Troak Village,
Pombeh Mon Commune,
Tbing District, Kampong
27/09/2020

ការពិនិត្យប្រជាជននៅភូមិព្រៃក្រវាញ ឃុំព្រៃក្រវាញ ស្រុកត្បូងឃ្មុំ

កាលបរិច្ឆេទ: ២៧ កញ្ញា ២០២០ ទីកន្លែង: ភូមិព្រៃក្រវាញ ឃុំព្រៃក្រវាញ ស្រុកត្បូងឃ្មុំ

ល.រ	ឈ្មោះ	តួនាទី	ស្ថាប័ន (ភូមិ)	ទូរស័ព្ទ	ហត្ថលេខា
01	គង់ហ៊ុន រ៉ាតា	ប្រធានប្រជាជន	គង់ហ៊ុន រ៉ាតា	011 655 292	
02	អ៊ា ច្បារីន	ប្រធានភូមិ	គ្រាហ៊ុន ច្បារីន	081 714143	
03	ឌីក វ៉ាន	ប្រធានការពារ	- - -	016 731530	
04	អ៊ា ក័រនេយាង	- - -	- - -	096 2740837	
05	ឌីន ចា	- - -	- - -		
06	ហាម ហ៊ាន	- - -	- - -	096 4773211	✓
07	សាត វ៉ាន	(ប្រធានការពារភូមិ)	- - -	093211432	✓

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Photographs of village level FPIC in Royes Village, Chhean Loeung Commune



Photographs of provincial level FPIC



Photographs of village level FPIC in Trapaing Troak village, PramBei Mom commune, Thbong district



Consultation with local authorities



Consultation with Chhean Loerng commune chief



Consultation with Royeas village chief